

FitTrack - AI

Deliver intelligent fitness tracking, workout planning, and personalized health insights

1. DEMONSTRATION FLOW

1. **Introduction:** Introduce FitTrack - AI and explain that it is a Node.js/Express REST backend designed for fitness activity tracking, user goal management, and AI-driven personalized health/workout insights.
2. **Project Structure:** Display the project folder structure in the code editor, highlighting controllers, models, routes, and middleware.
3. **Configuration:** Show environment configuration at a high level without exposing sensitive secret API keys.
4. **Server Startup:** Start MongoDB and launch the backend server using `npm run dev` or `npm start`. Demonstrate server startup and successful database connection in the terminal.
5. **API Client Setup:** Open Postman (or Insomnia) and import/use the dedicated FitTrack - AI API collection.
6. **User Registration:** Register a new user account (e.g., age, weight, height, fitness goals) and display the returned JWT.
7. **Authentication:** Log in with the registered user credentials to demonstrate successful authentication and token retrieval.
8. **User Profile:** Call the protected user profile endpoint using the Bearer JWT token.
9. **Activity / Workout Tracking:**
 - Post a new workout or daily activity log (e.g., running, cycling, gym workout) and show the returned activity record.
 - Retrieve the user's activity history and logged fitness metrics.
 - Update or delete a logged workout entry to demonstrate CRUD functionality.
10. **Public / General Fitness Endpoints:** Call the general fitness lookup endpoint (e.g., standard exercise database or calorie calculation endpoint) and show details (e.g., exercise type, intensity, target muscles, and fallback state).
11. **AI Workout / Health Insights:**
 - Call the protected AI Workout-Summary endpoint and display the AI-generated weekly workout breakdown.
 - Call the protected AI Health-Recommendation endpoint and show personalized diet, recovery, or routine recommendations generated based on user logs.
12. **Fallback Mechanism:** Briefly demonstrate the built-in fallback logic when external AI API keys or third-party fitness API keys are not configured.
13. **Conclusion:** Conclude the demo by summarizing JWT authentication, activity logging, AI insights generation, database persistence, and API testing.

2. DEMONSTRATION TALKING POINTS

- **Authentication & Security:** JWT is used to secure private endpoints and protect user health and activity data.
- **Database Management:** MongoDB and Mongoose models store user profiles, activity logs, daily goals, and fitness metrics.
- **Fitness Data Integration:** Exercise and workout data are processed via internal services or integrated third-party fitness APIs.
- **AI Engine Integration:** Gemini API processes user logs and fitness goals to deliver personalized workout routines and actionable health recommendations.
- **Resilience:** Fallback logic ensures key workout and AI endpoints remain fully testable even without active external API keys.

3. DEMO VIDEO LINK

Google Drive Demoo Video

Link: https://drive.google.com/file/d/15nWHqp19RJ8_lgYMZgoyTmSgu62cSHSw/view?usp=drive_link

Access note: Set the Google Drive sharing permission according to the SmartBridge submission requirement before submitting the repository.

4. SUBMISSION CHECKLIST

- ✓ All 8 phase-wise documents are added to the GitHub repository.
- ✓ Project source code is available in the repository.
- ✓ README and Postman collection are included.
- ✓ Sensitive API keys/secrets are not exposed in the repository.
- ✓ Final demo video link is added above.
- ✓ Important Postman/API screenshots are included if required by the submission format.